

Surag Nair

suragnair94@gmail.com

Education

Stanford University
PhD in Computer Science

SEP 2019 - DEC 2023

Stanford University
MS in Computer Science

SEP 2017 - JUN 2019

Cumulative Grade Point Average: 4.07

Indian Institute of Technology, Delhi
B.Tech in Electrical Engineering

JUL 2013 - JUN 2017

Cumulative Grade Point Average: 9.35/10

Employment

Genentech Inc.

JAN 2024 - PRESENT

Senior ML Scientist, South San Francisco

- Conducting research on building predictive models for deciphering the link between DNA sequences and diverse cellular/organismal phenotypes as part of the Biology Research AI Development (BRAID) team.

Stanford University

APR 2018 - DEC 2023

Graduate Research Assistant, Supervisor: Dr. Anshul Kundaje, Stanford

Developed and applied machine learning algorithms for regulatory genomics across multiple projects.

Mitigating spurious feature learning in conventional models of regulatory genomics

- Demonstrated that conventional machine learning models of regulatory genomics learn spurious features that result in inconsistent predictions, misleading feature interpretation, and erroneous sequence design.
- Exploring training design decisions that mitigate the learning of spurious sequence features.

Single-cell dissection of human skin cell reprogramming with DNA sequence models

- Analyzed single-cell RNA-seq and ATAC-seq of a time course of human skin cell reprogramming to iPSCs.
- Trained DNA sequence models to predict ATAC-seq at high resolution and performed model interpretation.
- Linked concentration of transcription factors and their DNA motif sequences to reprogramming progression.
- Proposed detailed mechanisms for how skin cells lose their identity within the first two days of reprogramming.

Speeding up In-silico Saturation Mutagenesis (ISM) for convolutional sequence models

- ISM is an interpretability method for deep learning sequence models in which each position in input sequence is perturbed and propagated through a trained model to measure the effect of the mutation on the output.
- Developed fastISM, an algorithm that speeds up ISM by over 10x for convolutional neural networks.

Cis-trans deep learning models for chromatin accessibility

- Developed models that incorporate DNA sequence with RNA expression data to predict chromatin accessibility.
- Demonstrated the ability to make predictions in unseen cell types and impute missing chromatin accessibility.
- Improved state-of-the-art by introducing new features, a ResNet model architecture, and training procedures.

Apple Inc.

JUN 2018 - AUG 2018

Software Engineering Intern, Siri International team, Cupertino

- Studied causes for failure in the multi-component machine learning pipeline for Siri, Apple's voice assistant.
- Devised and implemented an NLP pipeline for real-time detection of failure cases based on usage logs data.
- Workflow consisted of periodic PySpark and Python scripts running on terabytes of real-time data.

Selected Publications and Preprints

- **Transcription factor stoichiometry, motif affinity and syntax regulate single-cell chromatin dynamics during fibroblast reprogramming to pluripotency:** Surag Nair*, Mo Ameen*, Lakshman Sundaram, Anusri Pampari, Jacob Schreiber, Akshay Balsubramani, Will Wang, David Burns, Helen Blau, Ioannis Karakikes, Kevin Wang, Anshul Kundaje. *bioRxiv* 2023. 
- **The dynseq genome browser track displays context-specific sequence features at single-nucleotide resolution:** Surag Nair*, Arjun Barrett*, Daofeng Li*, Brian Raney, Brian Lee, Peter Kerpedjiev, Vivek Ramalingam, Anusri Pampari, Fritz Lekschas, Ting Wang, Maximilian Haeussler, Anshul Kundaje. *Nature Genetics* 2022.  

- **fastISM: Performant in-silico saturation mutagenesis for convolutional neural networks:** Surag Nair, Avanti Shrikumar, Jacob Schreiber, Anshul Kundaje. Bioinformatics 2022.  
- **Integrating regulatory DNA sequence and gene expression to predict genome-wide chromatin accessibility across cellular contexts:** Surag Nair*, Daniel Kim*, Jacob Perricone, Anshul Kundaje. Bioinformatics 2019.  
- **Single-cell multiome of the human retina and deep learning nominate causal variants in complex eye diseases:** Sean Wang, Surag Nair, Rui Li, Katerina Kraft, Anusri Pampari, Aman Patel, Joyce Kang, Christy Luoung, Anshul Kundaje, Howard Chang. Cell Genomics 2022.  
- **Accelerating in silico saturation mutagenesis using compressed sensing:** Jacob Schreiber, Surag Nair, Akshay Balsubramani, Anshul Kundaje. Bioinformatics 2022.  
- **AP-1 is a temporally regulated dual gatekeeper of reprogramming to pluripotency:** Glenn Markov, Thach Mai, Surag Nair, Anna Shcherbina, Yu Xin Wang, David Burns, Anshul Kundaje, Helen Blau. PNAS 2021. 
- **Deciphering regulatory DNA sequences and noncoding genetic variants using neural network models of massively parallel reporter assays:** Rajiv Movva, Peyton Greenside, Georgi K. Marinov, Surag Nair, Avanti Shrikumar, Anshul Kundaje. PLOS ONE 2019.  
- **Inferring Temporal Knowledge for Near-Periodic Recurrent Events:** Dinesh Raghu*, Surag Nair*, Mausam. International Joint Conference on Artificial Intelligence (IJCAI) 2018.  

**equal contribution*

Open Source Contributions

- **Multi-framework Alpha Zero**   [3000+ stars] WINTER 2017-18
· Developed a package for self-play based learning following the Alpha Zero paper by DeepMind. Allows easy addition of new games and works with all major deep learning frameworks (PyTorch, TensorFlow, Keras).
- **Stanford CS230 Deep Learning Starter Code**  [2000+ stars] WINTER 2017-18
· Developed starter code for deep learning projects in PyTorch with accompanying tutorials.
- **PyTorch Implementation of seqGAN Algorithm**  [600+ stars] AUTUMN 2017-18
· Implemented a LSTM based deep learning language model and trained it using a Generative Adversarial Network framework using Policy Gradients, in PyTorch, based on the paper by Lantao Yu et al. 2016.

Technical Skills

Languages : Python, R, C++, SQL
Softwares/Tools : PyTorch, TensorFlow, NumPy, PySpark

Invited/Selected Talks

- **Invited:** CEGS ICA Journal Club, NY Genome Center (Jan 2024): Motif syntax of fibroblast reprogramming
- **Invited:** IMSc Chennai (Aug 2023): Motif syntax of fibroblast reprogramming
- **Selected (Travel Fellowship):** ISMB (France, Jul 2023): Feature leakage in models of regulatory DNA ►
- **Invited:** IGVF Seminar Series (Apr 2023): Motif syntax of fibroblast reprogramming
- **Invited:** Bay Area Chromatin Club (Jul 2022): Motif syntax of fibroblast reprogramming
- **Selected:** Cold Spring Harbor Systems Biology (Mar 2022): Motif syntax of fibroblast reprogramming
- **Invited:** Genentech (Mar 2022): Motif syntax of fibroblast reprogramming
- **Selected:** ISMB 2021 (Jul 2021): Motif syntax of fibroblast reprogramming ►
- **Selected:** MLCB 2020 (Nov 2020): Speeding up in-silico saturation mutagenesis (fastISM) ►
- **Selected:** ISMB 2019 (Switzerland, Jul 2019): Cis-trans deep learning models for chromatin accessibility ►

Professional Activities

Journal Reviewer

- OUP Bioinformatics: 2019, 2023, 2024
- BMC Bioinformatics: 2022, 2023
- PLOS Computational Biology: 2022, 2023, 2024
- IEEE Transactions on Computational Biology and Bioinformatics: 2023
- Journal of Open Source Software: 2022, 2023

- Review Commons: 2021

Conference and Workshop Reviewer

- ICLR Machine Learning for Genomics Explorations (MLGenX): 2024
- Machine Learning in Computational Biology (MLCB): 2021, 2022, 2023
- NeurIPS AI4Science Workshop: 2023
- NeurIPS Generative AI and Biology (GenBio) Workshop: 2023
- ICML Workshop on Computational Biology: 2023
- International Conference on Intelligent Systems for Molecular Biology (ISMB): 2023

External Teaching

- Machine Learning in Health and Disease, International Centre for Theoretical Sciences, India: 2023
- Cold Spring Harbor Statistical Analysis of Genome Scale Data, Cold Spring Harbor, USA: 2022

Teaching Experience

CS230: Deep Learning

WINTER 2017-18

Course Instructors: Dr. Andrew Ng & Kian Katanforoosh, Stanford University 

- Project mentor for 14 teams who applied deep learning to domains including space imagery, translation, genomics, and photography. Developed questions for midterm examination. Graded exams and projects for 400+ students.

Scholastic Achievements

- 2017 Department Rank 5 (top 6%) among students of Electrical Engineering Department, Batch of 2017
- 2013 All India Rank 280 in IIT Joint Entrance Exam-Advanced out of over 1 million students
- 2013 National Top 40 in Indian National Chemistry Olympiad

Mentoring

- Areeb Gani (Montgomery Blair High School): 2022
- Arjun Barrett (The Harker School): 2021